cavity called **haemocoel**. Respiratory organs are gills, book gills, book lungs and trachea. Circulatory system is of open type. Sensory organs like antennae, eyes (compound and simple), statocysts (organs of balance/ equilibrium) are present. Excretion takes place through **malpighian tubules**, **green glands**, **coxal glands**, etc. They are mostly dioecious and oviparous; fertilization is usually internal. Development may be direct or indirect. Life history includes many larval stages followed by metamorphosis.

Examples : *Limulus* (King crab, a living fossil), *Palamnaeus* (Scorpion), *Eupagarus* (Hermit crab), *Apis* (Honey bee), *Musca* (House fly), Vectors- *Anopheles*, *Culex*, *Aedes* (mosquitoes), Economically important insects - *Apis*- (Honey bee), *Bombyx* (Silk worm), *Laccifer* (Lac insects), Gregarious pest - *Locusta* (Locust) (Figure 2.15)

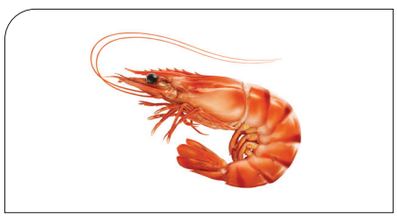


Spider silk is five times stronger than steel of the same diameter. It has been suggested that a Boeing 747 could be stopped in flight by a single pencil-width strand and spider silk is almost as strong as Kevlar, the toughest man-made polymer.

2.3.8 Phylum: Mollusca

(L. *molluscs* –soft bodied)

This is the second largest animal phylum. Molluscs are terrestrial and aquatic (marine or fresh water) and exhibit organ system level of body organisation. They are bilaterally symmetrical (except univalves eg. apple snail) triploblastic and coelomate animals. Body is covered by a calcareous shell and is unsegmented with a distinct head, muscular foot and a visceral hump or visceral mass. A soft layer of skin forms a mantle over the visceral hump. The space between the visceral mass and **mantle** (pallium) is called the mantle cavity in which a number of feather like gills (**ctenidia)** are present, which are respiratory in function. The digestive system is complete and mouth contains a rasping organ called **radula** with transverse rows of chitinous teeth for feeding (radula is absent in bivalves). The sense organs are tentacles, eyes and **osphraidium** (to test the purity of water and present in bivalves and gastropods). Excretory organs are nephridia. Open type of circulatory system is seen except for cephalopods such as squids, cuttle fishes and octopus. Blood contains **haemocyanin**, a copper containing


Prawn


Hermit crab


Locust


Scorpion


Spider


Limulus (Living fossil)

Figure 2.15 Examples of Arthropoda

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